Java Programming Lab (Classes and Objects)

1. Write a Java program to create a class Calculator that accepts two numbers and an operator as input. Using objects, perform the required arithmetic operation, handle division by zero, and maintain the history of the last five calculations.
2. Write a Java program to create a class Circle with radius as a data member. Using objects, calculate area and circumference, compare two circles based on area, and determine whether one circle can fit inside another.
3. Write a Java program to create a class Rectangle with length and breadth as data members. Using objects, calculate area and perimeter, determine whether the rectangle is a square, and compare two rectangles based on area and diagonal length.
4. Write a Java program to create a class Triangle with three sides as data members. Using objects, validate whether the sides form a triangle, identify its type (equilateral, isosceles, or scalene), and calculate its area using **Heron's Formula**.
5. Write a Java program to create a class Student with marks in multiple subjects as data members. Using objects, calculate total marks, percentage, grade, identify failed subjects, and determine class rank among multiple students.
6. Write a Java program to create a class Employee with basic salary, overtime hours, and leave deductions as data members. Using objects, calculate HRA, DA, tax deductions, bonuses, gross salary, and net salary according to salary slabs.
7. Write a Java program to create a class BankAccount with account number, balance, and transaction limit as data members. Using objects, perform deposit and withdrawal operations, apply penalties for insufficient balance, and calculate monthly interest on the remaining balance.
8. Write a Java program to create a class Interest with principal, rate, and time as data members. Using objects, calculate both simple interest and compound interest, compare the results, and display the difference in returns.
9. Write a Java program to create a class ElectricityBill with consumer type and units consumed as data members. Using objects, calculate the bill using slab rates, apply taxes, subsidies, and late payment charges, and generate the final payable amount.
10. Write a Java program to create a class Product with product ID, quantity, price per unit, expiry status, and discount category as data members. Using objects, calculate total cost, apply category-based discounts, add tax, and generate the final invoice.